OLS, IV/2SLS, and GMM in Matrix Form

Derivations Accompanying the Python Demo

These notes derive the three estimators implemented in the introductory Python demo: ordinary least squares (OLS), two-stage least squares (2SLS), and the generalized-method-of-moments (GMM) representation of 2SLS. The aim is to make explicit the linear algebra that the `statsmodels` and `linearmodels` packages execute internally, so that the hand-coded estimators in the notebook can be read line-by-line against the formulas below.

These notes are intended as intuitive guidance for the Python demo, not as a complete rigorous treatment. For the full assumptions and proofs, consult a standard econometrics textbook such as Hayashi *Econometrics (2011)*, Greene *Econometric Analysis (2012)*, or Cameron & Trivedi *Microeconometrics: Methods and Applications (2005)*. If you find any errors, please let me know.

NOT PART OF THE EXAM!

Contents

1 OLS in Matrix Form

Helpful notes with a more complete treatment can be found here: "OLS in Matrix Form".

1.1 The model and the least-squares criterion

Let $\mathbf{X}$ be an $n \times k$ matrix of regressors (one column is a vector of ones for the constant), let $\mathbf{y}$ be the $n \times 1$ vector of observations on the dependent variable, let $\boldsymbol{\varepsilon}$ be an $n \times 1$ vector of unobserved disturbances, and let $\boldsymbol{\beta}$ be the $k \times 1$ vector of unknown parameters. The model is

$$\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}. \tag{1}$$

The OLS estimator $\hat{\boldsymbol{\beta}}$ is defined as the minimizer of the sum of squared residuals. With the residual vector $\mathbf{e} = \mathbf{y} - \mathbf{X}\hat{\boldsymbol{\beta}}$, the criterion is

$$\mathbf{e}'\mathbf{e} = (\mathbf{y} - \mathbf{X}\hat{\boldsymbol{\beta}})'(\mathbf{y} - \mathbf{X}\hat{\boldsymbol{\beta}}) = \mathbf{y}'\mathbf{y} - 2\hat{\boldsymbol{\beta}}'\mathbf{X}'\mathbf{y} + \hat{\boldsymbol{\beta}}'\mathbf{X}'\mathbf{X}\hat{\boldsymbol{\beta}}, \tag{2}$$

where we used the fact that the transpose of a scalar is itself, so $\mathbf{y}'\mathbf{X}\hat{\boldsymbol{\beta}} = (\mathbf{y}'\mathbf{X}\hat{\boldsymbol{\beta}})' = \hat{\boldsymbol{\beta}}'\mathbf{X}'\mathbf{y}$.

1.2 The normal equations and the estimator

Differentiating (2) with respect to $\hat{\boldsymbol{\beta}}$ and setting the result to zero,

$$\frac{\partial\, \mathbf{e}'\mathbf{e}}{\partial \hat{\boldsymbol{\beta}}} = -2\mathbf{X}'\mathbf{y} + 2\mathbf{X}'\mathbf{X}\hat{\boldsymbol{\beta}} = \mathbf{0}, \tag{3}$$

which rearranges into the *normal equations*

$$(\mathbf{X}'\mathbf{X})\,\hat{\boldsymbol{\beta}} = \mathbf{X}'\mathbf{y}. \tag{4}$$

The matrix $\mathbf{X}'\mathbf{X}$ is square ($k \times k$) and symmetric. Provided $\mathbf{X}$ has full column rank, so that $(\mathbf{X}'\mathbf{X})^{-1}$ exists,[1] pre-multiplying both sides of (4) by $(\mathbf{X}'\mathbf{X})^{-1}$ gives

$$\boxed{\hat{\boldsymbol{\beta}} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{y}} \tag{5}$$

No assumptions about the disturbances are needed to reach (5); it is purely the algebra of the least-squares projection. In the notebook this is the single line `beta = inv(X.T @ X) @ (X.T @ y)`.

1.3 The variance of the OLS estimator

To characterize the sampling variability of $\hat{\boldsymbol{\beta}}$, substitute $\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}$ into (5):

$$\hat{\boldsymbol{\beta}} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'(\mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}) = \boldsymbol{\beta} + (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\boldsymbol{\varepsilon}. \tag{6}$$

Thus $\hat{\boldsymbol{\beta}}$ is unbiased whenever $\mathbb{E}[\mathbf{X}'\boldsymbol{\varepsilon}] = \mathbf{0}$ (e.g. $\mathbf{X}$ fixed, or $\mathbf{X}$ independent of $\boldsymbol{\varepsilon}$, with $\mathbb{E}[\boldsymbol{\varepsilon}] = \mathbf{0}$). For the variance, write the sampling error as $\hat{\boldsymbol{\beta}} - \boldsymbol{\beta} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\boldsymbol{\varepsilon}$, so

$$\operatorname{Var}(\hat{\boldsymbol{\beta}} \mid \mathbf{X}) = \mathbb{E}\Big[(\hat{\boldsymbol{\beta}} - \boldsymbol{\beta})(\hat{\boldsymbol{\beta}} - \boldsymbol{\beta})' \,\Big|\, \mathbf{X}\Big] = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\,\mathbb{E}[\boldsymbol{\varepsilon}\boldsymbol{\varepsilon}' \mid \mathbf{X}]\,\mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}. \tag{7}$$

[1] Full rank fails under perfect multicollinearity (a regressor is an exact linear combination of others) or when $n < k$. The second derivative of the criterion is $2\mathbf{X}'\mathbf{X}$, positive definite under full rank, confirming a minimum.

Under the homoskedasticity / no-autocorrelation assumption $\mathbb{E}[\boldsymbol{\varepsilon}\boldsymbol{\varepsilon}' \mid \mathbf{X}] = \sigma^2\mathbf{I}$, equation (7) collapses:

$$\begin{aligned}\operatorname{Var}(\hat{\boldsymbol{\beta}} \mid \mathbf{X}) &= (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'(\sigma^2\mathbf{I})\mathbf{X}(\mathbf{X}'\mathbf{X})^{-1} \\ &= \sigma^2(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{X}(\mathbf{X}'\mathbf{X})^{-1} \\ &= \sigma^2(\mathbf{X}'\mathbf{X})^{-1}. \end{aligned} \tag{8}$$

The disturbance variance is estimated by

$$\hat{\sigma}^2 = \frac{\mathbf{e}'\mathbf{e}}{n-k}, \tag{9}$$

the residual sum of squares divided by the degrees of freedom; the $n-k$ divisor makes $\hat{\sigma}^2$ unbiased.[2] The estimated variance–covariance matrix is therefore

$$\boxed{\widehat{\operatorname{Var}}(\hat{\boldsymbol{\beta}}) = \hat{\sigma}^2(\mathbf{X}'\mathbf{X})^{-1}} \tag{10}$$

and the standard errors are the square roots of its diagonal entries. This is exactly the `ols_formula` function in the notebook: form $(\mathbf{X}'\mathbf{X})^{-1}$, compute residuals, scale by $\hat{\sigma}^2$, and take $\sqrt{\operatorname{diag}(\cdot)}$.

Robust standard errors (aside). If instead one only assumes heteroskedasticity with no autocorrelation, so that $\mathbb{E}[\boldsymbol{\varepsilon}\boldsymbol{\varepsilon}' \mid \mathbf{X}] = \boldsymbol{\Omega} = \operatorname{diag}(\sigma_1^2, \ldots, \sigma_n^2)$, the middle of (7) no longer simplifies. The heteroskedasticity-consistent (White) estimator then replaces it with the empirical counterpart,

$$\widehat{\operatorname{Var}}_{\text{robust}}(\hat{\boldsymbol{\beta}}) = (\mathbf{X}'\mathbf{X})^{-1}\Big(\textstyle\sum_{i=1}^{n} \hat{e}_i^2\, \mathbf{x}_i\mathbf{x}_i'\Big)(\mathbf{X}'\mathbf{X})^{-1}. \tag{11}$$

This is the default reported by `linearmodels` for IV, and is the reason the demo must request `cov_type="unadjusted"` to recover the classical SEs of (10).

2 Instrumental Variables and Two-Stage Least Squares

2.1 Setup and the endogeneity problem

Partition the regressor matrix into exogenous and endogenous parts,

$$\mathbf{X} = \begin{bmatrix}\mathbf{X}_o & \mathbf{x}_e\end{bmatrix}, \tag{12}$$

where $\mathbf{X}_o$ ($n \times k_o$) collects the exogenous regressors (including the constant) and $\mathbf{x}_e$ ($n \times 1$) is the endogenous regressor (`educ` in the demo). OLS is inconsistent here because $\mathbf{x}_e$ is correlated with the disturbance: from (6), $\operatorname{plim}\hat{\boldsymbol{\beta}}_{\text{OLS}} = \boldsymbol{\beta} + \operatorname{plim}\big[(\mathbf{X}'\mathbf{X}/n)^{-1}(\mathbf{X}'\boldsymbol{\varepsilon}/n)\big] \neq \boldsymbol{\beta}$ whenever $\operatorname{plim}(\mathbf{X}'\boldsymbol{\varepsilon}/n) \neq \mathbf{0}$.

Let $\mathbf{Z}$ ($n \times \ell$) be the matrix of instruments. Crucially, $\mathbf{Z}$ contains *both* the excluded instrument(s) *and* the exogenous regressors $\mathbf{X}_o$, because the exogenous regressors are valid instruments for themselves. A valid instrument set satisfies

$$\text{relevance:} \quad \operatorname{plim}(\mathbf{Z}'\mathbf{X}/n) \text{ has full column rank,} \tag{13}$$

$$\text{exogeneity:} \quad \operatorname{plim}(\mathbf{Z}'\boldsymbol{\varepsilon}/n) = \mathbf{0}. \tag{14}$$

The *order condition* for identification is $\ell \geq k$: at least as many instruments as regressors. When $\ell = k$ the model is *just-identified* (the demo, with one excluded instrument `nearc4` for one endogenous regressor); when $\ell > k$ it is *over-identified*.

[2] Because $\mathbf{e} = \mathbf{M}\mathbf{y}$ with the idempotent residual-maker $\mathbf{M} = \mathbf{I} - \mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'$ of rank $n-k$, one has $\mathbb{E}[\mathbf{e}'\mathbf{e}] = \sigma^2(n-k)$.

2.2 2SLS as two regressions

The name "two-stage least squares" comes from the following two-step procedure.

Stage 1. Regress the endogenous regressor on the full instrument set and form fitted values. With the projection ("hat") matrix onto the column space of $\mathbf{Z}$,

$$\mathbf{P_Z} = \mathbf{Z}(\mathbf{Z}'\mathbf{Z})^{-1}\mathbf{Z}', \tag{15}$$

the fitted endogenous regressor is $\hat{\mathbf{x}}_e = \mathbf{P_Z}\mathbf{x}_e$. (The exogenous columns are unchanged by this projection, since $\mathbf{P_Z}\mathbf{X}_o = \mathbf{X}_o$: regressors that are already in $\mathbf{Z}$ are their own fitted values.) Stack the projected design matrix $\hat{\mathbf{X}} = \begin{bmatrix}\mathbf{X}_o & \hat{\mathbf{x}}_e\end{bmatrix} = \mathbf{P_Z}\mathbf{X}$.

Stage 2. Regress $\mathbf{y}$ on $\hat{\mathbf{X}}$ by OLS:

$$\hat{\boldsymbol{\beta}}_{2SLS} = (\hat{\mathbf{X}}'\hat{\mathbf{X}})^{-1}\hat{\mathbf{X}}'\mathbf{y}. \tag{16}$$

This is precisely the notebook's `two_sls_with_OLS`: run `ols_formula` once to get $\hat{\mathbf{x}}_e$, then run it again with $\hat{\mathbf{x}}_e$ in place of $\mathbf{x}_e$.

2.3 The compact projection formula

The two-step estimator (16) simplifies to a single expression. Using the *symmetry* ($\mathbf{P}'_\mathbf{Z} = \mathbf{P_Z}$) and *idempotency* ($\mathbf{P_Z}\mathbf{P_Z} = \mathbf{P_Z}$) of the projection matrix,

$$\hat{\mathbf{X}}'\hat{\mathbf{X}} = (\mathbf{P_Z}\mathbf{X})'(\mathbf{P_Z}\mathbf{X}) = \mathbf{X}'\mathbf{P}'_\mathbf{Z}\mathbf{P_Z}\mathbf{X} = \mathbf{X}'\mathbf{P_Z}\mathbf{X}, \qquad \hat{\mathbf{X}}'\mathbf{y} = \mathbf{X}'\mathbf{P}'_\mathbf{Z}\mathbf{y} = \mathbf{X}'\mathbf{P_Z}\mathbf{y}. \tag{17}$$

Substituting into (16),

$$\boxed{\hat{\boldsymbol{\beta}}_{2SLS} = (\mathbf{X}'\mathbf{P_Z}\mathbf{X})^{-1}\mathbf{X}'\mathbf{P_Z}\mathbf{y}} \tag{18}$$

This is the one-liner `two_sls_formula` in the notebook. Two special cases are worth noting. If $\ell = k$ (just-identified), $\mathbf{Z}'\mathbf{X}$ is square and invertible, and (18) reduces to the simple IV estimator

$$\hat{\boldsymbol{\beta}}_{IV} = (\mathbf{Z}'\mathbf{X})^{-1}\mathbf{Z}'\mathbf{y}. \tag{19}$$

If $\mathbf{Z} = \mathbf{X}$ (every regressor exogenous), then $\mathbf{P_Z}\mathbf{X} = \mathbf{X}$ and (18) collapses back to OLS in (5)—OLS is the special case of 2SLS with no endogeneity.

2.4 The variance of the 2SLS estimator

Substituting $\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}$ into (18) gives the sampling error

$$\hat{\boldsymbol{\beta}}_{2SLS} - \boldsymbol{\beta} = (\mathbf{X}'\mathbf{P_Z}\mathbf{X})^{-1}\mathbf{X}'\mathbf{P_Z}\,\boldsymbol{\varepsilon}. \tag{20}$$

Under the classical assumption $\mathbb{E}[\boldsymbol{\varepsilon}\boldsymbol{\varepsilon}' \mid \mathbf{Z}] = \sigma^2\mathbf{I}$, and using idempotency of $\mathbf{P_Z}$,

$$\begin{aligned}
\mathrm{Var}(\hat{\boldsymbol{\beta}}_{2SLS}) &= (\mathbf{X}'\mathbf{P_Z}\mathbf{X})^{-1}\mathbf{X}'\mathbf{P_Z}\,\mathbb{E}[\boldsymbol{\varepsilon}\boldsymbol{\varepsilon}']\,\mathbf{P_Z}\mathbf{X}(\mathbf{X}'\mathbf{P_Z}\mathbf{X})^{-1} \\
&= \sigma^2(\mathbf{X}'\mathbf{P_Z}\mathbf{X})^{-1}\mathbf{X}'\mathbf{P_Z}\mathbf{P_Z}\mathbf{X}(\mathbf{X}'\mathbf{P_Z}\mathbf{X})^{-1} \\
&= \sigma^2(\mathbf{X}'\mathbf{P_Z}\mathbf{X})^{-1}.
\end{aligned} \tag{21}$$

The estimator of σ^2 again uses the degrees-of-freedom correction,

$$\hat{\sigma}^2 = \frac{\hat{\boldsymbol{\varepsilon}}'\hat{\boldsymbol{\varepsilon}}}{n-k}, \qquad \hat{\boldsymbol{\varepsilon}} = \mathbf{y} - \mathbf{X}\hat{\boldsymbol{\beta}}_{2SLS}, \tag{22}$$

so that

$$\boxed{\widehat{\mathrm{Var}}(\hat{\boldsymbol{\beta}}_{2SLS}) = \hat{\sigma}^2(\mathbf{X}'\mathbf{P_Z}\mathbf{X})^{-1}} \tag{23}$$

As in the OLS case, (23) is the *homoskedastic* variance: it relies on $\mathbb{E}[\boldsymbol{\varepsilon}\boldsymbol{\varepsilon}' \mid \mathbf{Z}] = \sigma^2\mathbf{I}$ and should not be read as the general 2SLS variance. Under heteroskedasticity the middle term of the sandwich no longer collapses, and the heteroskedasticity-consistent estimator

$$\widehat{\mathrm{Var}}_{\mathrm{robust}}(\hat{\boldsymbol{\beta}}_{2SLS}) = (\mathbf{X}'\mathbf{P_Z}\mathbf{X})^{-1}\Big(\textstyle\sum_{i=1}^n \hat{e}_i^2\,\hat{\mathbf{x}}_i\hat{\mathbf{x}}_i'\Big)(\mathbf{X}'\mathbf{P_Z}\mathbf{X})^{-1}, \qquad \hat{\mathbf{x}}_i = \mathbf{P_Z}\text{-projected regressors,} \tag{24}$$

replaces it—the 2SLS analogue of (11). This is also why `linearmodels` reports robust SEs by default (cf. the `cov_type="unadjusted"` note above), and connects to the efficiency discussion in the GMM section below: (23) is the efficient-GMM variance *only* when the disturbances are homoskedastic.

The standard-error pitfall. The residuals in (22) must be formed with the *original* regressor matrix $\mathbf{X}$ (containing $\mathbf{x}_e$), *not* the projected matrix $\hat{\mathbf{X}}$ (containing $\hat{\mathbf{x}}_e$). If one naively reports the OLS standard errors from the second-stage regression of $\mathbf{y}$ on $\hat{\mathbf{X}}$, the implied residuals are $\mathbf{y} - \hat{\mathbf{X}}\hat{\boldsymbol{\beta}}_{2SLS}$, which use $\hat{\mathbf{x}}_e$ and understate the true residual variance—giving standard errors that are too small. This is exactly why `two_sls_with_OLS` in the notebook rebuilds the design matrix with the original `educ` column before computing $\hat{\sigma}^2$, and it is one of the central teaching points of the demo. (It is also why running two literal `regress` commands in Stata does not reproduce `ivregress`'s standard errors, whereas `ivregress` handles it internally.)

2.5 First stage and weak instruments

Relevance is testable from the first-stage regression of $\mathbf{x}_e$ on $\mathbf{Z}$. With a single excluded instrument the standard diagnostic is the first-stage F-statistic for the hypothesis that the excluded instrument has no effect; a common rule of thumb requires $F > 10$. In the just-identified case, weak-instrument-robust inference can be based on the Anderson–Rubin statistic.

3 The GMM Criterion Representation of 2SLS

3.1 Moment conditions

The exogeneity assumption is itself a statement about a population moment:

$$\mathbb{E}\big[\mathbf{Z}'(\mathbf{y} - \mathbf{X}\boldsymbol{\beta})\big] = \mathbf{0}. \tag{25}$$

That is, the instruments are orthogonal to the structural disturbance. The sample analogue is the $\ell \times 1$ vector of empirical moments

$$\mathbf{g}_n(\boldsymbol{\beta}) = \frac{1}{n}\,\mathbf{Z}'(\mathbf{y} - \mathbf{X}\boldsymbol{\beta}). \tag{26}$$

GMM chooses $\boldsymbol{\beta}$ to make these sample moments as close to zero as possible, in a metric defined by a symmetric positive-definite weighting matrix $\mathbf{W}$:

$$\hat{\boldsymbol{\beta}}_{GMM} = \arg\min_{\boldsymbol{\beta}}\ Q_n(\boldsymbol{\beta}), \qquad Q_n(\boldsymbol{\beta}) = \mathbf{g}_n(\boldsymbol{\beta})'\,\mathbf{W}\,\mathbf{g}_n(\boldsymbol{\beta}). \tag{27}$$

3.2 Just-identified case: the moment equations have an exact root

When $\ell = k$ (the demo's case), $\mathbf{g}_n(\boldsymbol{\beta})$ is a k-vector of equations in k unknowns, and we can set it *exactly* to zero regardless of $\mathbf{W}$:

$$\mathbf{Z}'(\mathbf{y} - \mathbf{X}\hat{\boldsymbol{\beta}}) = \mathbf{0} \implies \mathbf{Z}'\mathbf{X}\,\hat{\boldsymbol{\beta}} = \mathbf{Z}'\mathbf{y} \implies \hat{\boldsymbol{\beta}} = (\mathbf{Z}'\mathbf{X})^{-1}\mathbf{Z}'\mathbf{y}, \tag{28}$$

which is the simple IV estimator (19). Here the weighting matrix is irrelevant: any $\mathbf{W}$ gives the same root. This is why the notebook solves `gmm_moment_condition` $= \mathbf{Z}'(\mathbf{y} - \mathbf{X}\boldsymbol{\beta}) = \mathbf{0}$ directly with a root-finder, and why $\mathbf{W}$ never enters that computation.

3.3 Over-identified case: $\mathbf{W} = (\mathbf{Z}'\mathbf{Z})^{-1}$ delivers 2SLS

When $\ell > k$ there is no $\boldsymbol{\beta}$ that zeroes all ℓ moments simultaneously, so the choice of $\mathbf{W}$ matters. Minimizing (27) gives the first-order condition

$$\big(\mathbf{X}'\mathbf{Z}\big)\,\mathbf{W}\,\Big(\mathbf{Z}'(\mathbf{y} - \mathbf{X}\hat{\boldsymbol{\beta}})\Big) = \mathbf{0}, \tag{29}$$

which solves to the general linear-GMM estimator

$$\hat{\boldsymbol{\beta}}_{GMM} = \big(\mathbf{X}'\mathbf{Z}\,\mathbf{W}\,\mathbf{Z}'\mathbf{X}\big)^{-1}\,\mathbf{X}'\mathbf{Z}\,\mathbf{W}\,\mathbf{Z}'\mathbf{y}. \tag{30}$$

Choosing the 2SLS weighting matrix $\mathbf{W} = (\mathbf{Z}'\mathbf{Z})^{-1}$ and substituting,

$$\begin{aligned} \hat{\boldsymbol{\beta}}_{GMM} &= \big(\mathbf{X}'\mathbf{Z}(\mathbf{Z}'\mathbf{Z})^{-1}\mathbf{Z}'\mathbf{X}\big)^{-1}\,\mathbf{X}'\mathbf{Z}(\mathbf{Z}'\mathbf{Z})^{-1}\mathbf{Z}'\mathbf{y} \\ &= \big(\mathbf{X}'\mathbf{P}_{\mathbf{Z}}\mathbf{X}\big)^{-1}\,\mathbf{X}'\mathbf{P}_{\mathbf{Z}}\mathbf{y} \;=\; \hat{\boldsymbol{\beta}}_{2SLS}, \end{aligned} \tag{31}$$

since $\mathbf{Z}(\mathbf{Z}'\mathbf{Z})^{-1}\mathbf{Z}' = \mathbf{P}_{\mathbf{Z}}$. Thus **2SLS is the linear GMM estimator with weighting matrix** $(\mathbf{Z}'\mathbf{Z})^{-1}$—the unifying result the demo is built around.

3.4 Efficiency and the connection back to 2SLS

The *efficient* GMM estimator uses $\mathbf{W} = \mathbf{S}^{-1}$, where $\mathbf{S} = \mathrm{Var}\big(\frac{1}{\sqrt{n}}\mathbf{Z}'\boldsymbol{\varepsilon}\big)$ is the asymptotic variance of the sample moments. Under conditional homoskedasticity $\mathbb{E}[\boldsymbol{\varepsilon}\boldsymbol{\varepsilon}' \mid \mathbf{Z}] = \sigma^2\mathbf{I}$,

$$\mathbf{S} = \operatorname{plim}\frac{1}{n}\,\mathbf{Z}'\mathbb{E}[\boldsymbol{\varepsilon}\boldsymbol{\varepsilon}']\mathbf{Z} = \sigma^2\operatorname{plim}\frac{\mathbf{Z}'\mathbf{Z}}{n}, \tag{32}$$

so $\mathbf{S}^{-1} \propto (\mathbf{Z}'\mathbf{Z})^{-1}$ and efficient GMM coincides with 2SLS. Hence *2SLS is efficient GMM precisely when the disturbances are homoskedastic*; under heteroskedasticity or autocorrelation the efficient weight differs, which is the standard motivation for two-step or continuously-updated GMM. Over-identification additionally permits Hansen's J-test, $J = n\,\mathbf{g}_n(\hat{\boldsymbol{\beta}})'\mathbf{W}\mathbf{g}_n(\hat{\boldsymbol{\beta}}) \xrightarrow{d} \chi^2_{\ell-k}$, a joint test of the extra moment conditions (and thus a partial test of instrument validity); in the just-identified case $\ell - k = 0$ and there is nothing to test.

4 Summary of the Three Estimators

The three estimators implemented in the demo are one object viewed three ways:

$$\text{OLS:}\quad \hat{\boldsymbol{\beta}} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{y}, \tag{33}$$

$$\text{2SLS:}\quad \hat{\boldsymbol{\beta}} = (\mathbf{X}'\mathbf{P}_{\mathbf{Z}}\mathbf{X})^{-1}\mathbf{X}'\mathbf{P}_{\mathbf{Z}}\mathbf{y}, \qquad \mathbf{P}_{\mathbf{Z}} = \mathbf{Z}(\mathbf{Z}'\mathbf{Z})^{-1}\mathbf{Z}', \tag{34}$$

$$\text{GMM:}\quad \hat{\boldsymbol{\beta}} = \arg\min_{\boldsymbol{\beta}}\;\big[\mathbf{Z}'(\mathbf{y} - \mathbf{X}\boldsymbol{\beta})\big]'(\mathbf{Z}'\mathbf{Z})^{-1}\big[\mathbf{Z}'(\mathbf{y} - \mathbf{X}\boldsymbol{\beta})\big]. \tag{35}$$

OLS is 2SLS with $\mathbf{Z} = \mathbf{X}$; 2SLS is GMM with $\mathbf{W} = (\mathbf{Z}'\mathbf{Z})^{-1}$; and in the just-identified case all of them reduce to $(\mathbf{Z}'\mathbf{X})^{-1}\mathbf{Z}'\mathbf{y}$. The classical (homoskedastic) variance is $\hat{\sigma}^2(\mathbf{X}'\mathbf{X})^{-1}$ for OLS and $\hat{\sigma}^2(\mathbf{X}'\mathbf{P_Z}\mathbf{X})^{-1}$ for 2SLS, with $\hat{\sigma}^2$ divided by $n-k$ and residuals formed from the original $\mathbf{X}$.